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DCDA Capstone

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Links: https://github.com/marissahouff/DCDA40833-portfolio
https://marissahouff.github.io/DCDA40833-portfolio/

When I used Vibe coding in this lab I had two goals I wanted to achieve. First I wanted to add a dark/ light mode toggle to the top of my screen on the website. Second, I wanted to create a carousel for my labs on my home screen. In all honesty, I would not be able to write this code myself. These are features I have not learned and would not be able to come up with unless I was able to copy/paste. That being said, the AI was quite easy to use. I was able to understand why the code worked and would be able to write these codes now! I would still need to look at my example.

What I/m still unsure of is what aria-label/ aria-expanded does. After looking it up, I realized that it is basically a hidden description. It gives a text label to an element on a screen. It basically explains what a feature does. I modified lots of first generated content. For example, when I first asked for a dark/light mode toggle it had emojis that did not resonate with me. I was able to switch it to a clear toggle with a simple off/on mechanism. This looks much more cohesive with my website. Also, when I first had it build the carousel for my labs on the homepage, it did not work. At first it had 6 slides that showed the same lab in each stretched over 6 slides.This obviously made no sense on my website. After lots of back and forth with my AI, I was able to fix the issue and even added a photo feature to each slide. Each lab now has a better overall aesthetic as well as function.

When changing these features, I realized my hamburger icon in the top left corner had disappeared. I not only added it back but now when using it on larger screens it disappears. This is important because the hamburger feature is not needed on larger screens, as each drop down in the menu goes across the screen.

AI was most helpful when I was trying to build features that I genuinely did not know how to code from scratch. Adding a dark/light mode toggle and creating a lab carousel are not things we've directly learned in class, so having AI generate a starting point saved me hours of Googling random tutorials. I could focus on adjusting the design to match my website's aesthetic and functionality. It also simplified debugging when something didn't work, because I could paste the code back in and ask what was wrong.

AI was problematic when it produced code that technically worked but didn't make sense in my context. For example, like I mentioned previously, the first version of the carousel created six identical slides of the same lab, which clearly wasn't usable. It also added stylistic elements like emojis in the toggle that didn't fit my brand. In some cases, the code was more complex than necessary, and I had to simplify or restructure it. The AI didn't fully understand the layout of my site, which caused issues like my hamburger icon disappearing unexpectedly.

My role in this process was still essential. I had to evaluate whether the output made sense visually and functionally for my site. I decided what fit my aesthetic, what to remove, and what to modify. AI can generate code, but it can't understand my personal design choices, branding, or usability goals the way I can.

Vibe coding definitely has trade-offs. In this lab, it helped me build features that I realistically would not have attempted on my own yet. It pushed me outside my comfort zone and exposed me to new coding concepts. In that way, it helped my development because I was learning by modifying and analyzing real code. However, it could easily hinder growth if I just copy and pasted without reading or testing anything.

In terms of attribution, in an academic setting it's important to clearly state what the AI generated and what I modified. In a professional context, attribution might look different. I think AI tools are used without formal citations.

My personal guidelines for using AI in coding are that I will use it to generate starting points for features I don't yet know how to build, to debug errors after I've tried solving them myself, and to help explain concepts I don't understand. I won't use it to complete assignments without reading or modifying the code, and I won't claim it all as my work.